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1. **Purpose and Scope**

   This FMEA identifies how each user requirement in part 1 could fail, the effect and cause of each failure, the controls already in place, and the risk of each failure. The result decides how much testing each requirement needs. It follows a risk-based approach consistent with ISPE GAMP 5 (Second Edition) and the principles of ICH Q9 (R1). The intended use is the sterilization of wrapped instrument trays (LC-01) using cycle C-121, so failures can directly affect patient safety.

2. **Method**

   Each failure mode is scored 1 to 3 on three scales.

Score	Severity(S): Effect if it fails	Occurrence(O)chance it fails	Detection(D): Chance the failure goes unnoticed
3	Direct risk to patient safety, sterility assurance or personnel safety	Likely: Custom or complex function with no history	Not detected before harm: no alarm, check or later test
2	Product quality or data integrity impact, no direct patient risk	Possible: Configurable standard function	Detected later by a review, check, or other test
1	No direct impact on safety, quality, or data	Unlikely: Standard, vendor-tested feature	Detected immediately by an alarm, interlock or visible output

   Risk Priority Number (RPN) = S X O X D, from 1 to 27. High is 12 or more medium is 6 to 11 and low is 5 or less.

   Severity override rule: Any item with Severity 3 receives scripted testing even if its RPN is low. A low occurrence can hide a serious hazard, so the RPN alone is not enough.

Level	Testing approach
High	Scripted test with predefined steps, expected results, and recorded evidence, plus a challenge test that tries to make the function fail.
Medium	Scripted test with recorded results, with fewer challenge cases
Low	Unscripted (Exploratory) testing with a recorded outcome, or review of supplier documentation and factory test evidence

3. **FMEA table**

   Test references(OQ -xx, PQ, IQ) point to part 4 (OQ), the validation plan (Part 3) and the traceability matrix (part 5).

   3.1 General, Business workflow, security, regulatory, and integration requirements

URS ID	Failure Mode	Effect	Cause	Current Controls	S	O	D	RPN	Level	Test Action
URS-1.1	Documents not delivered	IQ cannot be completed	Supplier Oversight	Purchase order list deliverables	1	1	1	1	Low	IQ document review

URS-1.2	Ambient outside range	Control electronics fail, records lost	Room HVAC failure	Room monitoring	2	1	1	4	Low	IQ Datasheet and room record
URS-1.3	Unit too large for room	Installation delayed	Wrong drawing used	Site layout review	1	1	2	1	Low	IQ Measurement
URS-1.4	No service access	Maintenance and calibration missed	Poor layout	Layout review	1	1	1	1	Low	IQ Measurement
URS-1.5	Overweight Unit	Handling or structural problem	Weight not checked	Supplier Datasheet	2	1	1	2	Low	IQ Weight certificate
URS-1.6	Floor overload	Structural damage	Footprint miscalculated	Structural Review	2	1	1	2	Low	IQ calculation
URS-2.1	Load released without review	Non-sterile products could reach use	Release step does not enforce by the system	Procedure Only	3	2	2	12	High	OQ-04, with negative test
URS-2.2	Change or overdue requalification missed	Unqualified process used	No automatic flag	Procedure and Annual Schedule	3	2	2	18	High	OQ-15, with challenge
URS-2.3	Untrained operator runs a cycle	Operating error	Training status not checked by the system	Training records	2	2	2	8	Medium	OQ-14
URS-2.4	Load not linked to the cycle	Traceability lost	Manual entry skipped	Operator procedure	2	2	2	8	Medium	OQ-14
URS-2.5	Failed cycle released	Non-Sterile product	Deviation not linked to release	Procedure only	3	2	2	12	High	OQ-10, with challenge
URS-3.1	Unauthorized parameter change	Process drifts from the validated state	Roles not enforced	Role-based access	3	2	2	12	High	OQ-01, wi mggggg th negative test

URS-3.2	Weak password accepted	Unauthorized access	Password rules not enforce	Site IT policy	2	2	1	4	Low	OQ -02
URS-3.3	Lockout fails	Password guessing not stopped	Counter fault	Audit trail review	2	2	2	8	Medium	OQ-02
URS-3.4	Session left open	Action under another person's ID	No inactivity timeout	Operator Discipline	2	1	2	4	Low	OQ-02
URS-3.5	USB port enabled	Malware or unauthorized data copy	Ports not disabled	IT hardening	2	1	2	4	Low	Unscripted OQ check b
URS-4.1	Out-of-calibration probe	False pass of cycle	Overdue calibration unnoticed	Calibration schedule	3	2	2	12	High	IQ certificates and OQ-05
URS-4.2	Records incomplete	Inspection Finding	Report missing fields	Report review	2	2	2	8	Medium	OQ-07
URS-4.3	Unvalidated recipe used	Unproven process in production	Recipe state not enforced	Procedure only	3	2	2	12	High	OQ-15
URS-4.4	Signature incomplete	Record not valid	Required fields missing	Report review	2	2	2	8	Medium	OQ-04
URS-5.1	Clock Wrong	Unreliable time stamps	Time server failure	Time server Monitoring	2	2	2	8	Medium	OQ-16
URS-5.2	PDF export Fails		Network Fault	Alert shown to user	2	2	1	4	Low	Unscripted OQ check

3.2 Performance, Recovery, data Integrity, and non-functional requirements

URS ID	Failure Mode	Effect	Cause	Current Controls	S	O	D	RPN	Level	Test Action
URS-6.1	Chamber below 121.0 C during exposure	Under-processed load	Steam supply or control fault	Control loop and TT-101	3	2	2	12	High	OQ-6 and PQ

URS-6.1	Load point outside 121 to 124 C	Under-or over-processed load	Poor steam distribution	Thermal Mapping	3	2	2	12	High	OQ-06 and PQ
URS-6.1	Cold spots	Non-uniform heating	Air pockets, steam distribution	Thermal Mapping	3	2	2	12	High	OQ-06
URS-6.1	F0 too low at the cold point	Sterility not assured	Slow heat-up at the cold spot	F0 calculation	3	2	2	12	High	PQ
URS-6.1	Air trapped in the chamber or load	Steam cannot reach the load	Too few vacuum pluses	Air removal test	3	2	2	12	High	OQ-07
URS-6.1	Vacuum Leak	Poor steam contact	Door seal wear	Leak test	3	1	2	6	Medium	OQ-08
URS-6.1	Wet load	Sterile barrier compromised	Drying phases too short	Visual check	2	2	1	4	Low	PQ visual check
URS-6.1	Low temperature alarm fails	Failed cycle goes unnoticed	Alarm logic or wiring fault	Non independent of the system	3	2	3	18	High	OQ-09, with challenge
URS-6.2	Poor steam quality	Weak lethality, wet or superheated steam	Boiler carryover	Supplier steam control	2	2	2	8	Medium	Steam Quality test(Third party)
URS-6.3	Load does not fit the chamber	Wrong loading	Rack mismatch	Load drawing	2	1	1	2	Low	IQ and PQ layout check
URS-6.4	Cycle too long	Schedule impact	Phases run over its limit	HMI warning	1	2	1	2	Low	Timing recorded in OQ-06
URS-6.4	Stale display values	Operator misreads	Communication Stall	HMI watchdog	2	1	1	2	Low	Unscripted OQ check
URS-7.1	No backup	Record lost	Schedule failure	Black failure alarm	2	2	2	8	Medium	OQ-12
URS-7.2	Restore fails	Records unrecoverable	Corrupt backup	Checksum verification	2	2	3	12	High	OQ-12

URS-7.3	Power loss leaves chamber unsafe	Pressurized chamber, data loss	UPS or valve fault	UPS, fail-safe valves	3	2	2	12	High	OQ-11
URS-8.1	Data gaps	Cannot show cycle conditions	Logging stops	Logging Alarm	2	2	2	8	Medium	OQ-17
URS-8.2	Audit trail missing or editable	Changes hidden	Configuration error	Access control	3	2	3	18	High	OQ-03, with challenge
URS-8.3	Cycle record altered or deleted	Data integrity breach	Permissions left open	WORM behavior, backup	3	2	2	12	High	OQ-03, with challenge
URS-8.4	Date or time changed	False time stamps	Clock editable	Access control	2	2	2	8	Medium	OQ-16
URS-9.1	Door Opens under pressure	Injury	Interlock fault	Mechanical lock	3	1	1	3	Low(override applies)	OQ-13,because severity is 3
URS-9.2	Unclear alarms	Slow response	Raw codes shown	Alarm list review	2	1	1	2	Low	Checked in OQ-09
URS-9.3	Low availability	Schedule impact	Breakdowns	Maintenance records	1	2	1	2	Low	Review of operation records
URS-9.4	Maintenance missed	Reduced reliability	No schedule	Supplier PM schedule	2	1	2	4	Low	IQ document Review

4. Summary

Level	Count	Meaning for testing
High	16	Scripted test with challenge case for each
Medium	11	Scripted test with recorded result
Low	18	Unscripted testing or documentation review; URS-9.1 is scripted by the severity override

The highest RPN (18) belongs to requalification control, the low-temperature alarm, and the audit trail. In each case a failure could go unnoticed, which is why detection matters as much as severity.

5. Risk Control and re-assessment

- Every High item must have a passing scripted test, or a documented mitigation approved by QA, before the system is released for routine use.
- Any failed test is handled as a deviation and the related risk is re-rated.

- This FMEA is repeated after any significant change to hardware, software, recipes or intended use under change control.
- Low-risk items may rely on supplier factory and site acceptance test evidence, but only after that evidence has been reviewed and its use justified in the validation plan.

6. **Approvals**

Role	Name	Signature	Date
Prepared by (validation Engineer)			
Prepared by (Validation Manager)			
Approved by (QA)			